

Markscheme

Mathematics: analysis and approaches
Practice question bank

59. (a)

The general term of a sequence is defined by $u_n = 2n^2 + 3n - 7$.

Find the value of n for which $u_n = 145$.

set up the equation: $2n^2 + 3n - 7 = 145$ (M1)

$$2n^2 + 3n - 152 = 0 \quad \text{A1}$$

solve using the quadratic formula: $n = \frac{-3 \pm \sqrt{9 + 1216}}{4}$ (M1)

reject the negative, non-integer root; $n = 8$ R1

[4 marks]